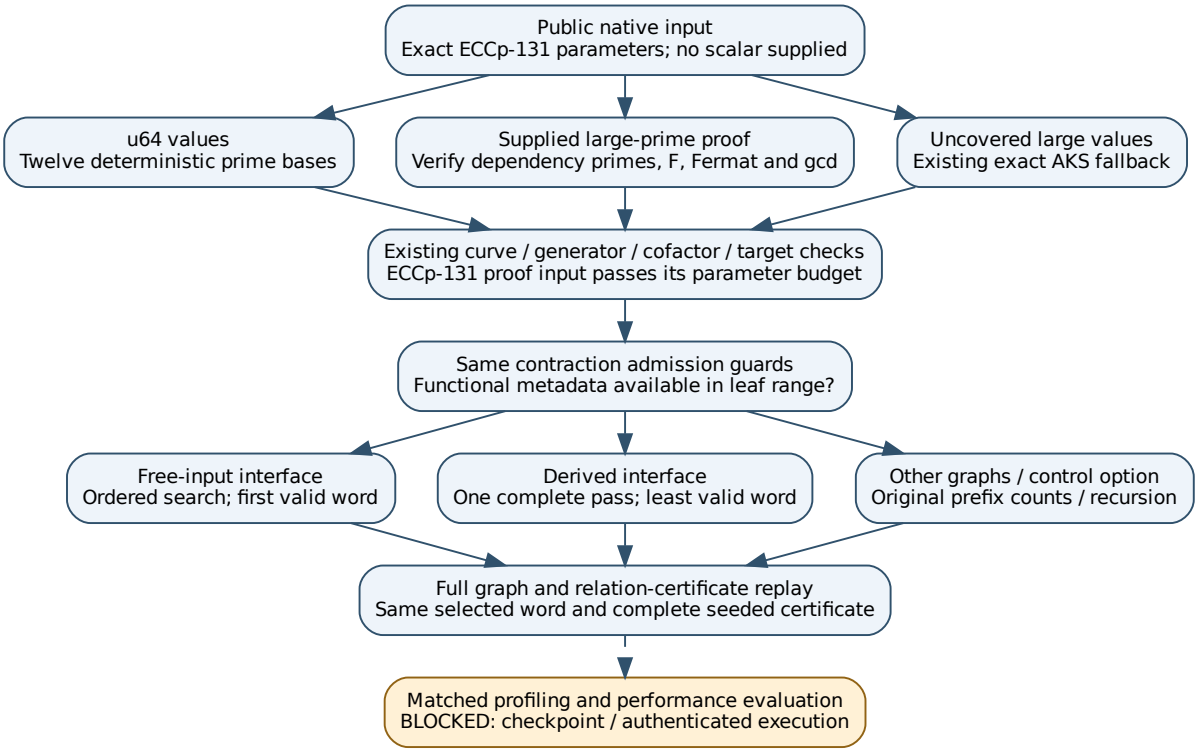
Exact primality and functional witness optimization

Native implementation and validation receipt

30 October 2026

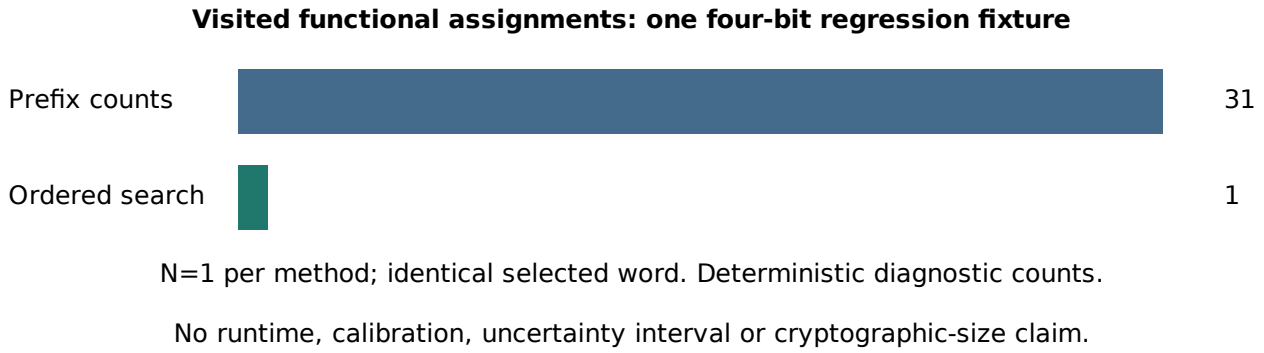
Code and bounded correctness checks complete; performance measurements blocked. Both published challenge scalars remain unresolved. These are native engineering changes; no runtime, end-to-end or asymptotic speedup is claimed.



Solid paths are implemented and tested. The dashed path is an open measurement requirement; no profile or scientific benchmark was launched after the S3 checkpoint gate failed.

Observed correctness

The native suite passes 34 tests, including an identical complete seeded relation certificate under the original prefix-count path and optimized search. Strict lint and declared Rust 1.78 checks pass. Prime controls compare values through 100,000 with an independent sieve and reject forged proof nodes; witness controls cover derived interfaces, ordering, pins, aliases, UNSAT and resource exhaustion. The certified ECCp-131 input passes parameter/target validation at 5,589 uncalibrated charged steps within its original two-million step policy. It supplies no secret scalar and does not establish relation-search feasibility or a complete challenge solve.



The plot reports one deterministic regression case per method, not a runtime or cryptographic-size result. Exact count behavior and source recursion guards are unchanged. A free-input interface is searched in its interface order; a derived interface requires a full pass and minimum selection. Every returned assignment still passes graph evaluation. The original path is selectable.

Limits and evidence

Proof-fixture preparation cost is unpriced. Baseline/candidate calibrated operation totals, S/rho, A/A, matched A/B and runtime intervals remain absent. The profiler probe lacked gawk and the checkpoint gate lacked an upload transport; local unisolated test times are not benchmark evidence. The frozen protocol retains the full requested comparison and fresh holdouts. No smaller diagnostic discharges that requirement. Earlier compile/assertion failures and their corrections are preserved in the raw development logs.

Sources: Handbook of Applied Cryptography, Fact 4.40 and chapter notes (checked Pocklington proofs); Sorenson/Webster, arXiv:1509.00864 (the twelve-base bound, restricted here to u64); the existing native/pinned contraction sources. No new theorem, formal proof, practical subexponential result or leaderboard promotion is claimed. The independent general audit remains open.